

PAPER_1410 — UQFF Derivation of Population III Initial Mass Function (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** G — Astrophysics (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_paradox({'paradox': 'pop3_imf'})` → $A_5 \times (D+1)/(D-1) = 100 \text{ M}_{\text{sun}}$ EXACT

Observation

First-generation (Pop III) stars are predicted to be top-heavy with characteristic mass $\sim 100 \text{ M}_{\text{sun}}$. UQFF derives this exact value.

UQFF Closed Identity

$$M_{\text{PopIII}} = A_5 \times (D_{\text{phys}} + 1) / (D_{\text{phys}} - 1) = 60 \times 5/3 = 100 \text{ M}_{\text{sun}} \text{ EXACT}$$

Physical Interpretation

Icosahedral group order $|A_5| = 60$ scales the structural mass scale; the $(D+1)/(D-1)$ factor encodes the spacetime-volume ratio of the early universe.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
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